AN ORDER-9 COUNTEREXAMPLE TO THE TRIPLICATION-TEMPLATE CONJECTURE

Abstract. Ogandzhanyants, Sadov, and Kondratieva conjectured that every triplication table obtained from a triplication template with an admissible key is induced by a strong starter. We give a keyed template of order 9 with an admissible key such that no strong starter in $\mathbb{Z}_{27}$ is congruous with the resulting table. A short carry identity over $\mathbb{F}_3$ certifies nonexistence.

We use the terminology and numbering of the published version of [1]. For a triplication table Σ_m, let $\mathrm{ST}(\Sigma_m)$ denote the set of ordered strong starters in $\mathbb{Z}_{3m}$ congruous with Σ_m. Conjecture 7.1 of [1] asserts that

$$t \in K(T_0, T_1, T_2) \quad \Longrightarrow \quad \mathrm{ST}\big(\Sigma_m(T_0, T_1, T_2, t)\big) \neq \varnothing.$$

Unsolvable triplication tables are already known: Theorem 7.8 of [1] exhibits three general triplication tables, of orders 9, 11 and 13, that are induced by no strong starter of order 27, 33, 39 respectively, and Remark 7.9 there reports 2625 unsolvable tables among 10^5 randomly generated tables of order 9. None of these is asserted to arise from a triplication template, and the order-9 table of [1, Table 14] does not: in no arrangement of its rows can a column be chosen that is a starter of $\mathbb{Z}_9$, so that table is not a keyed template for any key, and its equivalence class differs from that of (1). What is new below is thus not an unsolvable table of order 9, but one that lies in the template class of [1, Definition 4.2] with an admissible key, that is, in exactly the class quantified over by Conjecture 7.1.

Theorem 1. *In $\mathbb{Z}_9$, let*

$$\begin{aligned} T_0 &= [(1,2),(4,6),(5,8),(3,7)], \\ T_1 &= [(4,5),(1,3),(8,2),(8,3)], \\ T_2 &= [(5,6),(4,6),(7,1),(7,2)]. \end{aligned}$$

Then $\langle T_0, T_1, T_2\rangle$ is a triplication template, $1 \in K(T_0, T_1, T_2)$, and

$$\mathrm{ST}\big(\Sigma_9(T_0, T_1, T_2, 1)\big) = \varnothing.$$

Consequently, [1, Conjecture 7.1] *is false.*

Proof. The directed differences in each of T_0, T_1, T_2, in the displayed order, are $1, 2, 3, 4$ modulo 9. The pairing T_0 partitions $\mathbb{Z}_9^*$, while in the multiset

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union $T_1 \uplus T_2$ every element of $\mathbb{Z}_9^*$ occurs exactly twice. Thus $\langle T_0, T_1, T_2\rangle$ is a triplication template.

With key $t = 1$, the resulting keyed template is

$$\Sigma_9 = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & & (1,1) & \\ 1 & (1,2) & (5,6) & (6,7) \\ 2 & (4,6) & (2,4) & (5,7) \\ 3 & (5,8) & (0,3) & (8,2) \\ 4 & (3,7) & (0,4) & (8,3). \end{array} \tag{1}$$

Here 0 occurs twice and every nonzero residue occurs three times. The three directed differences in regular row r are all r, and the multiset of pair sums modulo 9 is

$$\{1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 6\}.$$

The special pair is $(1, 1)$, and the thirteen ordered pairs are distinct (in fact, their underlying unordered pairs are distinct). Hence (1) satisfies Definition 2.2 of [1]; equivalently, $1 \in K(T_0, T_1, T_2)$ by Definition 4.3.

Suppose that a strong starter in $\mathbb{Z}_{27}$ is congruous with (1). Index the thirteen table pairs by i and write the corresponding lifted pair as

$$x_i = u_i + 9U_i, \qquad y_i = v_i + 9V_i, \qquad U_i, V_i \in \{0, 1, 2\}.$$

Let $r_i \in \{0, 1, 2, 3, 4\}$ be the row index, with $r_i = 0$ for the special pair, and let $s_i \in \{0, \dots, 8\}$ be the residue of $u_i + v_i$ modulo 9. Define

$$\delta_i = \begin{cases} 1, & v_i - u_i < 0, \\ 0, & v_i - u_i \ge 0, \end{cases} \qquad \sigma_i = \begin{cases} 1, & u_i + v_i \ge 9, \\ 0, & u_i + v_i < 9. \end{cases}$$

Then

$$y_i - x_i \equiv r_i + 9(V_i - U_i - \delta_i) \pmod{27}, \tag{2}$$

$$x_i + y_i \equiv s_i + 9(U_i + V_i + \sigma_i) \pmod{27}. \tag{3}$$

All identities below are in $\mathbb{F}_3$.

Because the lifted pairs partition $\mathbb{Z}_{27}^*$, the three components above each nonzero residue modulo 9 have quotient digits $0, 1, 2$; the two components above residue 0 have quotient digits $1, 2$. Grouping by residue therefore gives

$$\sum_i (u_i U_i + v_i V_i) = 0. \tag{4}$$

For a fixed regular row r, put $D_i = V_i - U_i - \delta_i$. By (2), its three lifted directed differences are $r + 9D_i$. They represent distinct difference classes. Since $r \not\equiv -r \pmod 9$, two of them cannot be negatives of one another; hence the three values D_i are $0, 1, 2$. Summing over the regular rows yields

$$\sum_i r_i(V_i - U_i) = \sum_i r_i \delta_i. \tag{5}$$

For each $s \in \{1, 2, 3, 4\}$, exactly three pairs in (1) have sum s modulo 9. Their lifted sums are distinct, so by (3) the corresponding values $U_i + V_i + \sigma_i$

are $0, 1, 2$. The only other sum class is the singleton class $s = 6$, whose coefficient is zero in $\mathbb{F}_3$. Thus

$$\sum_i s_i(U_i + V_i) = -\sum_i s_i\sigma_i. \tag{6}$$

Add (4), (5), and (6). Since $r_i \equiv v_i - u_i$ and $s_i \equiv u_i + v_i$ modulo 9, the coefficients of U_i and V_i are respectively

$$u_i - r_i + s_i \equiv 3u_i = 0, \qquad v_i + r_i + s_i \equiv 3v_i = 0.$$

Every congruous strong starter must therefore satisfy the carry identity

$$\sum_i r_i\delta_i = \sum_i s_i\sigma_i. \tag{7}$$

In (1), the only pairs with negative ordinary directed difference are $(8, 2)$ in row 3 and $(8, 3)$ in row 4. Hence

$$\sum_i r_i\delta_i = 3 + 4 \equiv 1 \pmod 3.$$

The summation carries occur for

$$(5, 6), (6, 7), (4, 6), (5, 7), (5, 8), (8, 2), (3, 7), (8, 3),$$

whose sum residues are, respectively,

$$2, 4, 1, 3, 4, 1, 1, 2.$$

Their total is $18 \equiv 0 \pmod 3$, contradicting (7). Thus $\mathrm{ST}(\Sigma_9) = \varnothing$. □

Remark. The starter T_0 is near-strong of type I, whereas T_1 and T_2 are pseudostarters but not starters. Thus the example uses the pseudostarter generality of [1, Conjecture 7.1]. It leaves open the restriction to templates built entirely from starters and does not address Horton's conjecture.

References

[1] O. Ogandzhanyants, S. Sadov, and M. Kondratieva, *The triplication method for constructing strong starters*, J. Combin. Math. Combin. Comput. **131** (2026), 115–160. doi:10.61091/jcmcc131-08.